1. A company issues a bond with 5 years to maturity and an annual coupon of £100. The face value of the coupon is £1200. The annual continuously compounding risk-free interest rate is 5%.

   (a) What is the present value of the bond? Please round your answer to 3 decimal places. [4 marks]

   (b) What is the future value of the cash flow generated at six months before maturity? Please round your answer to 3 decimal places. [4 marks]

Solution

(a) The face value is discounted to

$$PV_{face} = 1200e^{-5 \cdot 0.05} = 934.561.$$

The coupon payments are discounted to

$$PV_{coupon} = 100(e^{-0.05} + e^{-2 \cdot 0.05} + \cdots + e^{-5 \cdot 0.05}) = 431.431$$

Therefore the present value is

$$PV = 934.561 + 431.431 = 1365.992.$$

(b) The future value of the cash flow is:

$$100(e^{0.5 \cdot 0.05} + e^{1.5 \cdot 0.05} + e^{2.5 \cdot 0.05} + e^{3.5 \cdot 0.05}) = 442.759.$$

2. The spot price of a stock S is £60, and it is expected to either increase by 10% or decrease by 5% every period over the next two 3-month periods. The annual continuously compounded risk-free interest rate is 4%.

   (a) i. Verify that this market is free of arbitrage. [2 marks]

   ii. Find the risk-neutral measure $\mathbb{Q}$ by identifying the values of q_u and q_d. [4 marks]

   iii. Draw a diagram to show the evolution of the stock price. [5 marks]

(b) Find the price of a 6-month European **call** option with the strike price £63, written on S. Please round your answer to 3 decimal places. [8 marks]

(c) Find the price of a 6-month American **put** option with the strike price £63, written on S. Please round your answer to 3 decimal places. Is an early exercise optimal? [13 marks]

(d) Let us introduce a new type of call option - the fixed-strike lookback call option. Given that the payoff of the lookback call, written on S, with strike price K at maturity T is

$$\max_{n=0,1,\cdots,T}\{S_n - K, 0\}.$$

Find the price of a 6-month lookback **call** option with the strike price £63, written on S. Please round your answer to 3 decimal places. [10 marks]

Solution

(a) i. We see that $d = 0.95$ and $u = 1.1$, and $e^{r\Delta t} = e^0.01 \approx 1.01$, and as

$$d < e^{r\Delta t} < u,$$

we conclude that the market is free of arbitrage.

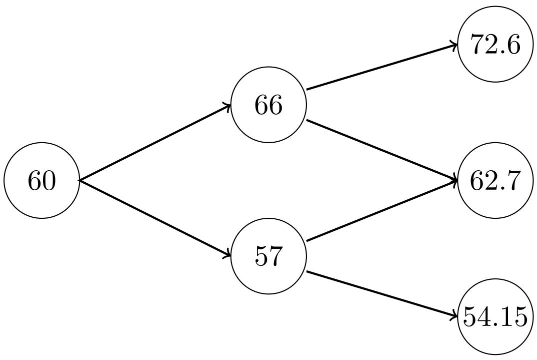
ii. Using the formulae introduced in lectures, the measure $\mathbb{Q}$ is determined by

$$q_u = \frac{e^{0.01} - 0.95}{1.1 - 0.95} \approx 0.4,$$

and

$$q_d = \frac{1.1 - e^{0.01}}{1.1 - 0.95} \approx 0.6.$$

iii.



(b) The values of European call are given by

$$\begin{aligned}
\Phi(u^2 S_0) &= \Phi(72.6) = (72.6 - 63)^+ = 9.6,\\
\Phi(udS_0) &= \Phi(62.7) = (62.7 - 63)^+ = 0,\\
\Phi(d^2 S_0) &= \Phi(54.15) = (54.15 - 63)^+ = 0.
\end{aligned}$$

Thus it follows that

$$\begin{aligned}
\Pi_0(\Phi(S_2)) &= e^{-2\cdot 0.01}\left(q_u^2 9.6 + 0 + 0\right)\\
&= 1.506
\end{aligned}$$

(c) At $n = 2$:

$$\begin{aligned}\Phi(72.6) &= (63 - 72.6)^+ = 0,\\ \Phi(62.7) &= (63 - 62.7)^+ = 0.3,\\ \Phi(54.15) &= (63 - 54.15)^+ = 8.85,\end{aligned}$$

At $n = 1$:

$$\begin{aligned}\Pi_1(\Phi)|_{S_1=66} &= \max\{(63 - 66)^+, e^{-0.01}(q_u\Phi(72.6) + q_d\Phi(62.7))\}\\ &= \max\{0, 0.178\}\\ &= 0.178, \qquad \text{no early exercise.}\end{aligned}$$

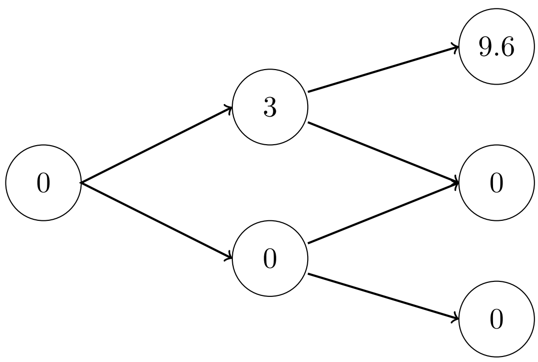
$$\begin{aligned}\Pi_1(\Phi)|_{S_1=57} &= \max\{(63 - 57)^+, e^{-0.01}(q_u\Phi(62.7) + q_d\Phi(54.15))\}\\ &= \max\{6, 5.376\}\\ &= 6, \qquad \text{exercise.}\end{aligned}$$

At $n = 0$:

$$\begin{aligned}\Pi_0(\Phi)|_{S_0=60} &= \max\{(63 - 60)^+, e^{-0.01}(q_u\Pi_1(\Phi)|_{S_1=66} + q_d\Pi_1(\Phi)|_{S_1=57})\}\\ &= \max\{3, 3.635\}\\ &= 3.635, \qquad \text{no early exercise.}\end{aligned}$$

The price of the option is £3.635, and an early exercise is optimal.

(d) The values of the lookback call are given by



$$\begin{aligned}\Phi(u^2S_0) &= \max\{0, 3, 9.6\} = 9.6,\\ \Phi(dS_1(u)) &= \max\{0, 3, 0\} = 3,\\ \Phi(uS_1(d)) &= \max\{0, 0, 0\} = 0,\\ \Phi(d^2S_0) &= \max\{0, 0, 0\} = 0,\end{aligned}$$

Thus we have

$$\Pi_0(\Phi(S_2)) = e^{-2\cdot 0.01}(q_u^2 9.6 + q_uq_d 3 + 0 + 0) = 2.211.$$

3. Let $W = (W_t)_{t\geq 0}$ be a standard Brownian motion on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

(a) Fix $T > 0$. Let $B = (B_t)_{0\leq t\leq T}$ be the stochastic process defined by

$$B_t = W_t - \frac{t}{T}W_T, \quad 0 \leq t \leq T.$$

This process is commonly known as the Brownian bridge. Find $\mathrm{Var}(B_t - B_s)$ for $0 \leq s \leq t \leq T$. Is B_t a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$? [8 marks]

(b) Fix $t \geq 0$. Consider the following stochastic differential equation for $s \geq t$:

$$\begin{cases} \mathrm{d}X_s = -\beta X_s \mathrm{d}s + \sqrt{2\theta}\mathrm{d}W_s, \\ X_t = x, \end{cases} \tag{1}$$

where $\beta, \theta > 0$, $x \in \mathbb{R}$ are fixed constants.

i. By applying Itô formula to $e^{\beta s}X_s$, solve the above SDE (1). [6 marks]

ii. Find $\mathbb{E}(X_s)$ and $\mathrm{Var}(X_s)$. [6 marks]

(c) Using the result from (b), find the solution to the partial differential equation

$$\frac{\partial u}{\partial t}(t,x) - x\frac{\partial u}{\partial x}(t,x) + \frac{\partial^2 u}{\partial x^2}(t,x) = 2u(t,x), \quad 0 \leq t \leq 1, x \in \mathbb{R}$$

subject to the terminal condition

$$u(1,x) = x^2.$$

[6 marks]

Solution

(a) We see that $\mathbb{E}(B_t - B_s) = 0$, and for $0 \leq s, t \leq T$,

$$\begin{aligned}\mathbb{E}(B_s B_t) &= \mathbb{E}(W_s W_t) - \frac{s}{T}\mathbb{E}(W_t W_T) - \frac{t}{T}\mathbb{E}(W_s W_T) + \frac{st}{T^2}\mathbb{E}(W_T^2) \\ &= s \wedge t - \frac{st}{T},\end{aligned}$$

Hence we have

$$\begin{aligned}\mathrm{Var}(B_t - B_s) &= \mathbb{E}((B_t - B_s)^2) \\ &= t - s - \frac{(t-s)^2}{T},\end{aligned}$$

therefore B_t is not a Brownian motion.

(b) i. Applying the Itô formula to $e^{\beta s}X_s$, then we have

$$\begin{aligned}\mathrm{d}(e^{\beta s}X_s) &= e^{\beta s}\mathrm{d}X_s + \beta e^{\beta s}X_s \mathrm{d}s \\ &= \sqrt{2\theta}e^{\beta s}\mathrm{d}W_s,\end{aligned}$$

Using the initial condition provided, it thus follows that

$$e^{\beta s}X_s - xe^{\beta t} = \sqrt{2\theta}\int_t^s e^{\beta\tau}\mathrm{d}W_\tau,$$

and thus

$$X_s = e^{-\beta s}\left(xe^{\beta t} + \sqrt{2\theta}\int_t^s e^{\beta\tau}\mathrm{d}W_\tau\right).$$

ii. We first notice that

$$\mathbb{E}(X_s) = xe^{-\beta(s-t)} + \sqrt{2\theta}e^{-\beta s}\mathbb{E}\left(\int_t^s e^{\beta\tau}\mathrm{d}W_\tau\right) = xe^{-\beta(s-t)},$$

and using Itô isometry, we have

$$\begin{aligned}\mathbb{E}(X_s^2) &= e^{-2\beta s}\mathbb{E}\left(x^2e^{2\beta t} + 2x\sqrt{2\theta}e^{\beta t}\int_t^s e^{\beta\tau}\mathrm{d}W_\tau + 2\theta\left(\int_t^s e^{\beta\tau}\mathrm{d}W_\tau\right)^2\right)\\ &= e^{-2\beta s}\left(x^2e^{2\beta t} + 2\theta\int_t^s e^{2\beta\tau}\mathrm{d}\tau\right)\\ &= x^2e^{-2\beta(s-t)} + \frac{\theta}{\beta}(1-e^{-2\beta(s-t)}).\end{aligned}$$

Therefore, we conclude

$$\mathrm{Var}(X_s) = \mathbb{E}(X_s^2) - \mathbb{E}(X_s)^2 = \frac{\theta}{\beta}(1-e^{-2\beta(s-t)}).$$

(c) By the Feynman-Kac formula, the solution to the given PDE can be expressed as an expectation:

$$u(t,x) = e^{-2(1-t)}\mathbb{E}(X_1^2),$$

where X satisfies for $0 \le t \le s$

$$\begin{cases}\mathrm{d}X_s = -X_s\mathrm{d}s + \sqrt{2}\mathrm{d}W_s,\\ X_t = x.\end{cases}$$

Hence using the computation in (b), we have

$$u(t,x) = e^{-2(1-t)}\left(x^2e^{-2(1-t)} + (1-e^{-2(1-t)})\right).$$

4. Suppose that $T > 0$ and $(W_t)_{t\in[0,T]}$ is a standard Brownian motion. The price of a stock at time t is assumed to be a geometric Brownian motion

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma S_t\mathrm{d}W_t$$

for $0 \le t \le T$, and r the continuously compounding risk-free interest rate per year and $\sigma > 0$ is the volatility. $S_0 > 0$ is the spot price of the stock.

(a) Let $\alpha > 0$ be some fixed constant and set $X_t = S_t^\alpha$ for every $t \in [0,T]$. Find the stochastic differential equation satisfied by X_t. [6 marks]

(b) The European digital call option written on the non-dividend paying stock S with strike price $K > 0$ has the following pay-off function at maturity time T:

$$f(S_T) = 1_{\{S_T \ge K\}} = \begin{cases} 1, & S_T \ge K, \\ 0, & S_T < K. \end{cases}$$

The European digital put option written on the non-dividend paying stock S with strike price $K > 0$ has the following pay-off function at maturity time T:

$$f(S_T) = 1_{\{S_T < K\}} = \begin{cases} 1, & S_T < K, \\ 0, & S_T \ge K. \end{cases}$$

i. Let $N(x)$ be the cumulative distribution function of the standard normal distribution $N(0,1)$. Using $N(x)$, find the Black-Scholes prices $\Pi_0(\mathrm{Call}(T,K))$ and $\Pi_0(\mathrm{Put}(T,K))$ of the European digital call and put respectively. You must show all intermediate steps, including how to find the probability and expectation needed here. [14 marks]

ii. By comparing the portfolio consisting of a long digital call and a long digital put with a suitable zero coupon bond, establish a Put-Call Parity equation for European digital options. [4 marks]

Solution

(a) Applying the Itô formula to X_t, we obtain that

$$\mathrm{d}X_t = \alpha S_t^{\alpha-1}\mathrm{d}S_t + \frac{1}{2}\alpha(\alpha-1)S_t^{\alpha-2}\mathrm{d}\langle S\rangle_t,$$

Using the SDE satisfied by S_t, we compute that the quadratic variation

$$\mathrm{d}\langle S\rangle_t = \sigma^2 S_t^2\mathrm{d}t$$

Thus it follows that

$$\begin{aligned}
\mathrm{d}X_t &= \alpha S_t^{\alpha-1}(rS_t\mathrm{d}t + \sigma S_t\mathrm{d}W_t) + \frac{1}{2}\alpha(\alpha-1)S_t^{\alpha-2}\sigma^2 S_t^2\mathrm{d}t \\
&= \alpha r S_t^\alpha\mathrm{d}t + \alpha\sigma S_t^\alpha\mathrm{d}W_t + \frac{1}{2}\alpha(\alpha-1)\sigma^2 S_t^\alpha\mathrm{d}t \\
&= \left(\alpha r + \frac{1}{2}\alpha(\alpha-1)\sigma^2\right)X_t\mathrm{d}t + \alpha\sigma X_t\mathrm{d}W_t.
\end{aligned}$$

Therefore, the SDE satisfied by X_t is

$$\begin{aligned}
\mathrm{d}X_t &= \left(\alpha r + \frac{1}{2}\alpha(\alpha-1)\sigma^2\right)X_t\mathrm{d}t + \alpha\sigma X_t\mathrm{d}W_t, \\
X_0 &= S_0^\alpha.
\end{aligned}$$

(b) i. First notice that $S_t = S_0 e^{(r-\frac{\sigma^2}{2})t+\sigma W_t}$. Using the given pay-off function we have

$$\begin{aligned}\Pi_0(Call(T,K)) &= e^{-rT}\mathbb{E}(1_{\{S_T\geq K\}})\\ &= e^{-rT}\mathbb{P}(S_T\geq K).\end{aligned}$$

Now we compute that

$$\begin{aligned}\mathbb{P}(S_T\geq K) &= \mathbb{P}(S_0 e^{(r-\frac{\sigma^2}{2})T+\sigma W_T} > K)\\ &= \mathbb{P}\left(W_T \geq \frac{\ln(\frac{K}{S_0}) - (r-\frac{\sigma^2}{2})T}{\sigma}\right)\\ &= \mathbb{P}\left(Z \geq -\frac{\ln(\frac{S_0}{K}) + (r-\frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right)\\ &= \mathbb{P}\left(Z < \frac{\ln(\frac{S_0}{K}) + (r-\frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right)\\ &= N(d_-).\end{aligned}$$

Therefore, we get the Black-Scholes price of the digital call option is

$$\Pi_0(\mathrm{Call}(T,K)) = e^{-rT}N(d_-).$$

Similar to the previous part, we have

$$\begin{aligned}\Pi_0(Put(T,K)) &= e^{-rT}\mathbb{E}(1_{\{S_T<K\}})\\ &= e^{-rT}\mathbb{P}(S_T<K)\\ &= e^{-rT}\mathbb{P}(Z<-d_-))\\ &= e^{-rT}N(-d_-).\end{aligned}$$

ii. Consider two portfolios:
Portfolio A: a long digital call and a long digital put,
Portfolio B: a zero coupon bond.
We notice that the value of Portfolio A at maturity is always 1, thus, if we set the face value of the bond in B is also 1, then using the Law of One price, we get

$$\Pi_0(\mathrm{Call}(T,K)) + \Pi_0(\mathrm{Put}(T,K)) = e^{-rT}.$$